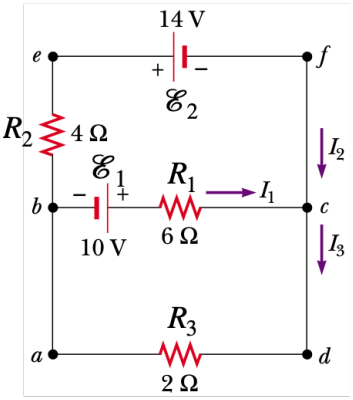
Kirchhoff Loop Example



To the right is a circuit with two batteries and some loops. You will use Kirchhoff's rules to analyze the circuit, but first answer the following fill-in-the-blank questions. *1 point each.*

1. Kirchhoff's junction rule is a mathematical expression of conservation of **charge**.

2. Kirchhoff's loop rule is a mathematical expression of conservation of **energy**.

3. Also write down an expression relating the currents I_1, I_2 and I_3 using the "junction rule". What junction (e.g., junction b) are you considering when you write this? *1 point.*

   At junction c we write $I_1 + I_2 = I_3$. At junction b we would write the (mathematically equivalent) $I_3 = I_1 + I_2$.

4. Using the terms R_1, R_2, R_3, I_1, I_2, I_3, $\mathscr{E}_1$ and $\mathscr{E}_2$ write down "loop rule" expressions for at least two different loops. State explicitly what the loop is by giving the order of its corners (e.g. *abcda*). *5 points.*

   For the loop 'aefda' we have
   $$-I_2R_2 - \mathscr{E}_2 - I_3R_3 = 0.$$
   For the loop 'abcda' we have
   $$\mathscr{E}_1 - I_1R_1 - I_3R_3 = 0.$$
   For the loop 'befcb' we have
   $$-I_2R_2 - \mathscr{E}_2 + I_1R_1 - \mathscr{E}_1 = 0.$$
   Note that subtracting loop 'abcda' from loop 'aefda' gives loop 'befcb'. So, the three loops are not linearly independent.

5. What are the numerical values of the three currents I_1, I_2, and I_3? **Show your work**. (*Note:* You may use Wolfram Alpha or another online calculator to check your math. But for full credit, show how you eliminate variables to get down to equations with only one unknown.) *8 points.*

We have three equations for three unknowns (I_1, I_2, and I_3),

$$I_3 = I_1 + I_2, \qquad -I_2R_2 - \mathscr{E}_2 - I_3R_3 = 0, \quad \text{and} \quad \mathscr{E}_1 - I_1R_1 - I_3R_3 = 0.$$

Let's use the junction condition to remove I_3 from the two loop equations

$$-I_2R_2 - \mathscr{E}_2 - (I_1 + I_2)R_3 = 0, \qquad \mathscr{E}_1 - I_1R_1 - (I_1 + I_2)R_3 = 0,$$

or

$$-I_2(R_2 + R_3) - \mathscr{E}_2 - I_1R_3 = 0, \qquad \mathscr{E}_1 - I_1(R_1 + R_3) - I_2R_3 = 0.$$

In this case we will benefit from plugging in for the resistances and emfs, so

$$-I_2 \cdot 6\ \Omega - 14\ \text{V} - I_1 \cdot 2\ \Omega = 0, \qquad 10\ \text{V} - I_1 \cdot 8\ \Omega - I_2 \cdot 2\ \Omega = 0.$$

Multiply the second equation by -3 and get

$$-30\ \text{V} + I_1 \cdot 24\ \Omega + I_2 \cdot 6\ \Omega = 0.$$

Add it to the first equation to get

$$-44\ \text{V} + I_1 \cdot 22\ \Omega = 0, \qquad \Longrightarrow \qquad I_1 = 2\ \text{A}.$$

Plug this into second equation so,

$$10\ \text{V} - 2\ \text{A} \cdot 8\ \Omega - I_2 \cdot 2\ \Omega = 0 \qquad \Longrightarrow \qquad -6\ \text{V} - I_2 \cdot 2\ \Omega = 0 \qquad \Longrightarrow \qquad I_2 = -3\ \text{A}.$$

Finally, from the junction condition we have

$$I_3 = 2\ \text{A} - 3\ \text{A} = -1\ \text{A}.$$